AspeQt-2k26

High-Performance Peripheral Emulator for Atari 8-Bit Computers

Introduction High-Speed SIO Disk & Folder TNFS Client Modem Bridge Wiring Diagram
Raspberry Pi Time Sync Printer (P:) Web Pipe (W:)

Introduction

AspeQt-2k26 is a modernized version of the classic AspeQt emulator, updated to use the Qt6 framework. It transforms your modern PC into a powerful peripheral server for Atari 8-bit computers (400/800/XL/XE).

Using a standard SIO2PC cable (USB or Serial), AspeQt can emulate:

- **Disk Drives (D1:-D8:)** - Supports ATR, XFD, PRO images, and direct Folder Mounting.
- **TNFS Client** - Mount and stream disk images directly from the internet (FujiNet protocol).
- **Modem Bridge** - Connect physical Atari modems to Telnet BBSs via the PC.
- **Printers (P:)** - Captures printed output to text or graphical view.
- **Clipboard Bridge (Y:)** - Bi-directional copy/paste between PC and Atari.
- **Web Pipe (W:)** - Multi-channel gateway for Python scripts and AI.

High-Speed SIO

AspeQt supports "divisor 0" high-speed SIO (approx. 126kbps) with custom OS ROMs (like QMEG or Hiassoft) and standard high-speed protocols (UltraSpeed, Happy, etc.).

> **Note:** Reliable high-speed operation depends on your SIO2PC hardware. FTDI-based USB cables generally offer the best performance.

Disk & Folder Usage

Mounting Disk Images

Drag and drop `.atr`, `.xfd`, or `.pro` files directly into the drive slots (D1-D8). Double-clicking a slot opens the file browser.

Virtual DOS (Folder Images)

You can mount a local PC directory as a writable disk. To boot from a folder, right-click the slot and select **Mount Folder**.

TNFS Client

AspeQt-2k26 includes native support for TNFS (The Network File System), allowing you to mount and stream disk images directly from the internet using the FujiNet protocol. Access this via the **Tools** menu.

Modem Bridge (Internet Adapter)

[cite_start]

The AspeQt-2k26 Modem Bridge is a hardware-software hybrid solution that allows vintage Atari 8-bit computers to access modern Telnet BBSs and TCP/IP services[cite: 5]. [cite_start]Unlike pure software emulators (like the R: Device), this feature utilizes the physical Atari 850 Interface (or compatible RS-232 hardware like the P:R: Connection or ICD R-Verter)[cite: 6]. [cite_start]AspeQt acts as a "Virtual Hayes Modem," bridging the physical serial data from the Atari to a TCP/IP socket on the PC[cite: 7]. [cite_start]This allows the use of authentic vintage terminal software (BobTerm, Ice-T, Express!) to connect to the internet[cite: 8].

1. The "Dual-Cable" Topology

[cite_start]

To utilize the Modem Bridge, your setup requires two distinct connections to the PC[cite: 10].

[cite_start]
- **Disk I/O Path:** Atari SIO -> SIO2PC -> PC USB (COM A)[cite: 13]. [cite_start]Handles D1: through D8:[cite: 14].

  [cite_start]

- **Modem Path:** Atari 850 (Port 1) -> Null Modem -> PC USB (COM B)[cite: 15]. [cite_start]Handles R: (The Modem Bridge)[cite: 15].

```
[ ATARI COMPUTER ]                                     [ PC / ASPEQT ]
+----------------+                                 +---------------------+
|                |            (1)                  |                     |
|   SIO PORT     |=======[ SIO2PC ]===============>|   [ USB PORT 1 ]    |
|                |                                 |   (SIO / Disks)     |
+----------------+                                 +---------------------+
        |
        | (Daisy Chain SIO Cable)
        |
+----------------+                                 +---------------------+
|  850 INTERFACE |            (2)                  |                     |
|   (R: DEVICE)  |======[ USB-SERIAL ]============>|   [ USB PORT 2 ]    |
|     PORT 1     |        ADAPTER                  |   (Modem Bridge)    |
+----------------+      (Null Modem)               +---------------------+
                                                              |
                                                              v
                                                      ( TO INTERNET )
```

2. Configuration Reference

[cite_start]

Configure the bridge settings via Tools > Options > Modem Bridge[cite: 61]. [cite_start]Correct settings are vital for stable communication[cite: 61].

Setting	Description & Recommended Value
Enable Modem Bridge	Master switch for the feature. When checked, AspeQt opens the serial port and asserts DTR/RTS signals. [cite_start]Note: If unchecked, the Atari will see the modem as "Offline" or "Not Ready"[cite: 62].
Serial Port	Select the COM port (Windows) or TTY device (Linux/Mac) corresponding to your USB-to-Serial adapter. [cite_start]Do NOT select your SIO2PC adapter here[cite: 62].
Baud Rate	The communication speed between PC and Atari. Must match your terminal software settings. [cite_start]Recommended: 19200 (Standard for Atari 850)[cite: 62].

Flow Control	Enables hardware RTS/CTS handshaking. Recommended: OFF (Unchecked) unless you have a high-quality 5-wire Null Modem cable. [cite_start]If enabled with a 3-wire cable, the connection will freeze[cite: 62].
Local Echo	If checked, AspeQt echoes all typed characters back to the Atari. Use this if you cannot see what you are typing in your terminal software. [cite_start]Recommended: OFF (Most BBS software handles echoing remotely)[cite: 62].
Phonebook File	Path to the phonebook.xml file storing your BBS list. [cite_start]Defaults to the application directory[cite: 62].

3. Using the Phonebook

[cite_start]

The Phonebook is essential for managing BBS addresses and storing credentials for the macro system[cite: 64].

- [cite_start]
  Access: Tools -> Phonebook[cite: 65].
- [cite_start]
  Add Entry: Click Add[cite: 66]. [cite_start]Fill in Name, Address, Port, and optional Login/Password to use the Auto-Type macros[cite: 67, 68, 69, 70].
- [cite_start]
  Dialing: Select an entry and click Dial[cite: 71]. [cite_start]AspeQt will send the ATDT string to the Atari automatically[cite: 72].

4. Hayes Emulation & Command Set

[cite_start]

The bridge emulates a standard Hayes-compatible modem[cite: 75]. [cite_start]You can issue these commands from any Atari terminal program[cite: 75].

Command / Key	Description
`AT`	Attention. Returns OK. [cite_start]Used to verify connection[cite: 77].

`ATDT <Name>`	Dial by Name. Example: ATDT Basement. [cite_start]Looks up IP/Port in Phonebook and loads credentials[cite: 77].
`ATDT <IP:Port>`	Raw Dial. Example: ATDT 192.168.1.5:23. [cite_start]Connects directly (No macros available)[cite: 77].
`ATH`	Hangup. [cite_start]Disconnects the TCP session[cite: 77].
`+++`	Escape. Pauses data mode and switches to command mode. [cite_start]Requires 1 second silence before/after[cite: 77].
`$ESC+U$`	Auto-User. [cite_start]Types the Username saved in the current Phonebook entry[cite: 80].
`$ESC+P$`	Auto-Pass. [cite_start]Types the Password saved in the current Phonebook entry[cite: 80].
`$ESC+H$`	Panic Hangup. [cite_start]Immediately terminates the connection and forces Command Mode[cite: 80].

5. ATASCII Translation

[cite_start]

Modern servers expect ASCII[cite: 82]. [cite_start]The bridge automatically performs these translations[cite: 82]:

- [cite_start]
  Delete Key: Atari Delete (126/127) is converted to ASCII Backspace (0x08)[cite: 83].
- [cite_start]
  EOL: Atari EOL (155) is converted to CR/LF[cite: 84].

6. Troubleshooting

- [cite_start]
  "Modem Not Responding" in Terminal: Cause: The Atari 850 does not detect the DSR/CD signal[cite: 87]. [cite_start]Fix: Verify the wiring in Section 3. Ensure AspeQt "Enable Modem Bridge" is checked[cite: 88].

  [cite_start]

- **"Garbage Characters" on Screen:** Cause: Baud rate mismatch[cite: 90]. [cite_start]Fix: Ensure the baud rate in AspeQt Options matches the baud rate configured in your Atari terminal software (e.g., 19200)[cite: 91].

  [cite_start]
- **"Session Frozen / Hanging":** Cause: Telnet flow control deadlock[cite: 93]. [cite_start]Fix: Press ESC + H to force a disconnect, then type ATZ to reset the bridge[cite: 94].

Appendix: Corrected Wiring for Atari 850 Null Modem Cable

[cite_start]

Standard off-the-shelf Null Modem cables will not work with the Atari 850 without modification[cite: 40]. [cite_start]The 850 requires the DSR (Data Set Ready) and CD (Carrier Detect) pins to be held HIGH to detect the presence of a modem[cite: 41]. [cite_start]AspeQt asserts the PC's DTR (Data Terminal Ready) signal when the bridge is active[cite: 42]. [cite_start]You must construct a cable that routes this DTR signal to the 850's DSR/CD pins[cite: 43].

The "Magic" Signal: Loopbacks

[cite_start]

On the PC side: Bridge Pin 1 (CD) and Pin 6 (DSR) together, and connect them to the wire coming from the Atari's Pin 1 (DTR)[cite: 47]. [cite_start]This ensures the PC knows the Atari is ready[cite: 48].

[cite_start]

On the Atari side: Bridge Pin 2 (CD) and Pin 6 (DSR) together, and connect them to the wire coming from the PC's Pin 4 (DTR)[cite: 48]. [cite_start]This ensures the Atari 850 detects a valid carrier tone and opens the port for data[cite: 49].

```
ATARI 850 (PORT 1)                              PC USB ADAPTER
[ DB9 MALE Connector ]                          [ DB9 FEMALE Connector ]
(Plugs into Atari 850)                          (Plugs into PC)

  PIN   FUNCTION                                  FUNCTION   PIN
 +---+------------+                              +------------+---+
 | 1 |  DTR (Out) |----------------------------->| CD / DSR   |1+6|
 +---+------------+          (Cross)             +------------+---+
 |6+2|  DSR / CD  |<-----------------------------| DTR (Out)  | 4 |
 +---+------------+                              +------------+---+
```

```
    +---+------------+                              +------------+---+
    | 3 |  TX  (Out) |----------------------------->| RX  (In)   | 2 |
    +---+------------+          (Cross)             +------------+---+
    | 4 |  RX  (In)  |<-----------------------------| TX  (Out)  | 3 |
    +---+------------+                              +------------+---+

    +---+------------+                              +------------+---+
    | 7 |  RTS (Out) |----------------------------->| CTS (In)   | 8 |
    +---+------------+          (Cross)             +------------+---+
    | 8 |  CTS (In)  |<-----------------------------| RTS (Out)  | 7 |
    +---+------------+                              +------------+---+

    +---+------------+                              +------------+---+
    | 5 |  GROUND    |------------------------------| GROUND     | 5 |
    +---+------------+          (Direct)            +------------+---+
```

Raspberry Pi Implementation (Direct UART)

[cite_start]

This section details the configuration for users running AspeQt-2k26 directly on a Raspberry Pi[cite: 96]. [cite_start]In this configuration, the Raspberry Pi acts as both the host for the emulator and the physical modem interface[cite: 97].

8.1 Hardware Architecture (Pi Edition)

[cite_start]

Unlike the PC setup which requires two USB cables, this setup utilizes the Pi's built-in GPIO pins for the modem connection[cite: 99].

- [cite_start]
  Disk I/O Path: Atari SIO → SIO2PC Raspberry Pi USB Port[cite: 100]. [cite_start]Function: Handles standard disk drives (D1:-D8:)[cite: 100].
- [cite_start]
  Modem Path: Atari 850 (Port 1) Level Shifter → Raspberry Pi GPIO Header[cite: 101]. [cite_start]Function: Handles R: (The Modem Bridge)[cite: 101].

> [cite_start]**CRITICAL Voltage Hazard:** The Atari 850 uses RS-232 voltage levels (±12V)[cite: 103]. [cite_start]The Raspberry Pi GPIO uses 3.3V TTL logic[cite: 103]. [cite_start]DO NOT connect the Atari 850 directly to the Raspberry Pi[cite: 104]. [cite_start]YOU MUST use a 3.3V RS-232 to TTL Level Shifter (e.g.,

> MAX3232) to bridge the connection[cite: 105]. [cite_start]Direct connection will permanently damage the Pi[cite: 106].

8.2 Wiring: Pi GPIO to Atari 850 (Port 1)

[cite_start]

Since the Raspberry Pi UART lacks standard DTR/DSR control pins, we utilize a "Local Loopback" on the Atari side to trick the 850 into detecting a "Ready" modem[cite: 108]. [cite_start]Inside the DB9 connector plugging into the Atari 850, solder a small jumper wire connecting Pin 1 (DTR) to both Pin 6 (DSR) and Pin 2 (CD)[cite: 111, 112]. [cite_start]This ensures the Atari 850 instantly sees the "modem" (Pi) as online when the Atari is powered on[cite: 113].

[cite_start] [cite_start] [cite_start] [cite_start] [cite_start] [cite_start] [cite_start] [cite_start]

Signal Function	Atari 850 (DB9 Pin)	Direction	Level Shifter (TTL Side)	Raspberry Pi GPIO
Transmit Data	Pin 3 (TX)	Atari → Pi	T2IN → T2OUT	GPIO 15 (RXD) - Pin 10 [cite: 110]
Receive Data	Pin 4 (RX)	Pi → Atari	R2OUT → R2IN	GPIO 14 (TXD) - Pin 8 [cite: 110]
Ground	Pin 5 (GND)	Common	GND	GND - Pin 6 [cite: 110]
Clear To Send	Pin 8 (CTS)	Pi → Atari	T1OUT → T1IN	GPIO 17 (RTS) - Pin 11 [cite: 110]
Request To Send	Pin 7 (RTS)	Atari → Pi	R1OUT → R1IN	GPIO 16 (CTS) - Pin 36 [cite: 110]
Modem Ready	Pin 6 (DSR)	Loopback	--	-- [cite: 110]
Carrier Detect	Pin 2 (CD)	Loopback	--	-- [cite: 110]
Terminal Ready	Pin 1 (DTR)	Loopback	--	-- [cite: 110]

8.3 Raspberry Pi OS Configuration

[cite_start]

Before running AspeQt, you must free the UART from the Linux console[cite: 115].

[cite_start]
1. **Disable Serial Console:** Run `sudo raspi-config`[cite: 117]. [cite_start]Navigate to Interface Options -> Serial Port[cite: 118, 119]. [cite_start]Answer "No" to "login shell to be accessible over serial?"[cite: 121, 122]. [cite_start]Answer "Yes" to "serial port hardware to be enabled?"[cite: 124].

[cite_start]
2. **Optimize UART (Disable Bluetooth):** To ensure stable timing for the Atari 850, use the primary UART (PL011) by disabling Bluetooth[cite: 126]. [cite_start]Edit config: `sudo nano /boot/config.txt` [cite: 127] [cite_start]and add line: `dtoverlay=disable-bt`[cite: 128]. [cite_start]Save and Reboot[cite: 129].

AspeQt Configuration (Pi Specifics)

[cite_start]

Change the settings in Tools > Options > Modem Bridge to match the Linux environment[cite: 131]:

[cite_start]
- **Serial Port:** /dev/ttyAMA0 (if BT disabled) OR /dev/serial0 [cite: 138]

[cite_start]
- **Baud Rate:** 19200 (Recommended) [cite: 139]
- **Flow Control:** OFF (Recommended for initial setup). [cite_start]Only enable if the "Loopback" wiring above is 100% verified[cite: 140].

Time Synchronization (Real-Time Clock)

AspeQt-2k26 allows your Atari 8-bit computer to read the current date and time from your PC. This is essential for file timestamps in DOS.

Recommended Method: ApeTime Protocol

AspeQt natively emulates the **ApeTime** device. This is the modern standard supported by most DOS variants.

- **SpartaDOS X:** Enable the `DATE` and `TIME` drivers, or load `ATIME.SYS`.
- **RealDOS / BW-DOS:** Load the `ATIME.SYS` driver in your `CONFIG.SYS`.
- **MyDOS:** Requires a specialized autorun utility.

> **Note:** You do **not** need to run any special client software on the PC side. AspeQt handles the protocol automatically in the background.

Printer Emulation (P:)

Output sent to `P:` is captured by AspeQt. View the output via **Tools > View Printer Output**.

The Y: Device (Clipboard Bridge)

The **Y: Device** bridges your Atari 8-bit and your modern PC's system clipboard. It allows you to "Paste" code from your PC directly into Atari BASIC, or "Copy" listings from the Atari back to your PC.

- **Import Code (PC → Atari):** Copy text on PC, then type `ENTER "Y:"` on Atari.
- **Export Code (Atari → PC):** Type `LIST "Y:"` on Atari, then Paste on PC.

The W: Device (Web Pipe)

The **W:** device transforms the Atari 8-bit into a "Thin Client" capable of interacting with modern web services. It acts as a transparent proxy, forwarding standard BASIC text commands to a backend script running on your PC via HTTP.

1. Multi-Channel Support (W1: - W4:)

AspeQt-2k26 supports up to **4 simultaneous connections**. This allows for complex "Full Duplex" applications where one channel reads data while another writes data.

- **W:** or **W1:** - Primary Channel (Device $57)
- **W2:** - Secondary Channel (Device $56)
- **W3:** - Tertiary Channel (Device $55)
- **W4:** - Quaternary Channel (Device $54)

```
10 REM -- QUAD CHANNEL EXAMPLE --
20 OPEN #1,8,0,"W1:http://192.168.1.5:5000/cmd"  :REM Command Channel
```

```
30 OPEN #2,4,0,"W2:http://192.168.1.5:5000/data" :REM Data Stream
40 OPEN #3,4,0,"W3:http://192.168.1.5:5000/status":REM Background Poll
50 PRINT #1;"GET STATUS"
60 INPUT #3,STATUS$
70 IF STATUS$="READY" THEN INPUT #2,DATA$
80 CLOSE #1 : CLOSE #2 : CLOSE #3
```

2. Configuration: Text vs. Binary Mode

The W: device can be configured globally via **Options > Emulation**, or overridden per-connection using the command parameters.

- **Text Mode:** Best for BASIC and Web APIs.
  - *Write:* Converts Atari EOL (`$9B`) → Unix Newline (`\n`).
  - *Read:* Converts Unix Newline (`\n`) → Atari EOL (`$9B`).
- **Binary Mode:** Best for Game Loaders and Asset Streaming.
  - Data is passed raw, byte-for-byte. No translation occurs.

3. Application Examples Library

This library contains a collection of "Hybrid Computing" examples:

1. Gemini AI Chatbot

Turns the Atari into a client for Google's Gemini Large Language Model. You can type natural language questions and receive summarized answers on the Atari screen.

- **BASIC File:** GEMINI_AI.LST
- **Python Script:** gemini_ai.py
- **Dependencies:** `pip install google-generativeai`

2. Basic Graphical Weather Station

Fetches live weather data from the Open-Meteo API and displays it on the Atari using GRAPHICS 5.

- **BASIC File:** WEATHER.LST
- **Python Script:** weather.py
- **Dependencies:** `pip install requests`

3. Cloud Notepad

Demonstrates Reading and Writing to the PC's hard drive. It acts as a "Cloud Drive" for text.

- **BASIC File:** CLOUD_NOTE.LST
- **Python Script:** cloud_note.py

4. Text-to-Speech Synthesizer

The "Talking Atari". Any text typed on the Atari is sent to the PC, where it is read aloud using the Linux espeak engine.

- **BASIC File:** TTS_SERVER.LST
- **Python Script:** tts_server.py

5. Web News Scraper

Connects to a news website, scrapes HTML, and delivers a plain-text headline summary to the Atari.

- **BASIC File:** SCRAPER.LST
- **Python Script:** scraper.py

6. Remote Execution (Linux & Windows)

Allows the Atari to execute shell commands or launch GUI applications on the Host PC.

- **BASIC Files:** EXEC_LINUX.LST / EXEC_WIN.LST
- **Python Scripts:** exec_linux.py / exec_win.py

7. File Upload / Telemetry

A utility to upload raw data or text files from the Atari to the PC.

- **BASIC File:** FILE_UPLOAD.LST
- **Python Script:** file_upload.py